

Muhammad Subhan

Email: subhanmughal931@gmail.com WhatsApp: 0344-8261148
Portfolio: m-subhan.vercel.app GitHub: github.com/subhanmughal1

SUMMARY

Python & AI Automation Engineer skilled in FastAPI backend development, biometric hardware integrations, AI-powered RAG systems, scalable REST APIs, AI chatbots, and autonomous agent workflows using n8n. Experienced in integrating ZKTeco biometric systems, vector databases, and LLM-based applications (LangChain, LangGraph — Learning) for real-world environments.

TECHNICAL SKILLS

- **Frontend** – HTML, CSS, Tailwind CSS, Responsive Design
- **Backend** – Python, Flask, FastAPI, REST API, API Integration, Biometric Hardware Integration (Fingerprint, Face Recognition, Palm, RFID Card)
- **AI / LLM** – RAG Pipelines, Vector Databases, LangChain, LangGraph, Claude AI, OpenAI APIs
- **Automation** – n8n (Drag & Drop Automation), Workflow Automation, AI Agent Pipelines
- **Database** – SQLite3, PostgreSQL, Redis
- **Version Control** – Git, GitHub (Branching, Pull Requests, Code Reviews, Collaborative Workflows)
- **Tools** – Postman, Visual Studio Code, Windsurf Code, Claude AI

WORK EXPERIENCE

Full Stack Developer | Mozzine Technologies (On-site) October 2025 – Present

- Built desktop biometric attendance application using Python and PyQt6 with MVC architecture, integrating ZKTeco devices and SQLite3 for attendance data processing.
- Built secure FastAPI-based REST APIs for biometric device communication, authentication, and data synchronization.
- Integrated and supported multiple ZKTeco biometric device models — fingerprint, face recognition, palm recognition, and RFID card readers.
- Delivered direct biometric customer support: diagnosed and resolved fingerprint/face recognition errors, device connectivity, enrollment, synchronization failures, and deployment issues for clients.
- Built n8n workflow automations for operational task streamlining and data pipeline orchestration.
- Implemented multi-threaded data sync workers (QThread) for device communication without UI blocking.

PROJECTS

AI RAG Agents & Chatbots Python, FastAPI, Claude AI, n8n, Vector DB

- Customer Support RAG Agent — automated query resolution via LLM + vector search over knowledge base documents with n8n pipeline triggers.
- Document Q&A RAG Agent — users upload PDFs/docs and query them in natural language for internal policy and manual lookups.
- Product Knowledge RAG Bot — trained on product catalogs & FAQs to handle pre-sales and technical queries without human intervention.
- Multi-Source RAG Agent — aggregates data from databases, APIs & documents to answer complex business queries in real-time.
- General-purpose AI chatbots with conversation memory deployed across web and WhatsApp channels.

n8n Workflow Automation n8n, Python, REST APIs, Webhooks

- Lead Capture & Data Sync — automated form submissions to CRM, welcome emails, and data synchronization between Google Sheets, Airtable, and PostgreSQL using scheduled triggers and webhooks.
- Notification & Alert System — created n8n workflows to monitor API endpoints and send real-time Slack/email alerts on failures or threshold breaches.

Python Automation

- Medical & Healthcare Info Bot — RAG agent providing accurate health and medical information from trusted knowledge base sources.
- CSS (Central Superior Services) & PPSC Exams Preparation RAG Agent — intelligent chatbot helping students prepare for CSS and PPSC exams with topic-wise Q&A and past paper analysis.
- Developed automation scripts using Python and n8n to streamline repetitive tasks and improve workflow efficiency.

EDUCATION

Bachelor of Science in Computer Science – University of Narowal (UON) **Nov 2021 – May 2025**

Intermediate (Pre-Engineering) – Royal College of Science, Narowal **May 2018 – Oct 2020**